

Mg₂Fe₂O₅ Nanoparticle-Decorated Ca₂Fe₂O₅–CaFe₂O₄ Heterostructure for Efficient Photocatalytic CO₂ Conversion

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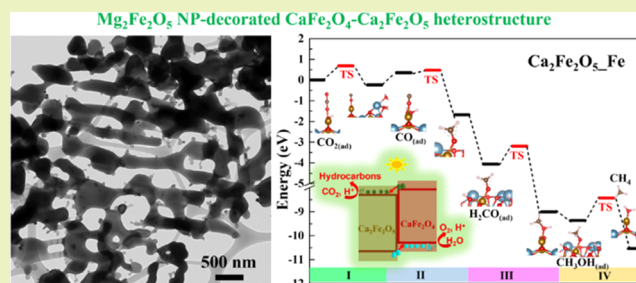
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ABSTRACT: In this work, a CaFe₂O₄ porous network and a Mg₂Fe₂O₅ nanoparticle (NP)-decorated Ca₂Fe₂O₅–CaFe₂O₄ heterostructure are synthesized using the solution combustion method. Ca₂Fe₂O₅ and Mg₂Fe₂O₅ NPs are incorporated into the CaFe₂O₄ network based on density functional theory (DFT) calculations for enhancing the CO₂ adsorption of the heterostructure. With the addition of Ca₂Fe₂O₅ and Mg₂Fe₂O₅ NPs, this CaFe₂O₄-based heterostructure demonstrates significantly improved photocatalytic activity for CO₂ conversion compared to the pristine CaFe₂O₄ network. CH₄ and CH₃CHO are produced, and there is no H₂ detected from the photocatalytic conversion of CO₂ and H₂O over the Mg₂Fe₂O₅ NP-decorated Ca₂Fe₂O₅–CaFe₂O₄ heterostructure. Selectivities of 81.7 and 18.3%, respectively, for CH₄ and CH₃CHO are achieved in this process. DFT calculations indicate that among the three components in the heterostructure, Ca₂Fe₂O₅ is the active site with the lowest activation energy for the conversion of CO₂ and H₂O to CH₄. Type-II charge transfer dynamics is suggested to take place in the staggered CaFe₂O₄–Ca₂Fe₂O₅ heterojunction to improve charge separation, which allows the photoelectrons to be well collected on the Ca₂Fe₂O₅ side for the reduction of CO₂. Accordingly, the CO₂ adsorption, charge separation, and surface CO₂ conversion efficiencies are all enhanced with the incorporation of Ca₂Fe₂O₅ and Mg₂Fe₂O₅ into the CaFe₂O₄ porous network, resulting in the boosted photocatalytic activity for CO₂ conversion in the Mg₂Fe₂O₅ NP-decorated CaFe₂O₄–Ca₂Fe₂O₅ heterostructure.

KEYWORDS: Ca₂Fe₂O₅–CaFe₂O₄ heterostructure, photocatalytic CO₂ conversion, density functional theory calculation, CO₂ adsorption site, conversion active site



INTRODUCTION

Calcium ferrite (CaFe₂O₄) comprising earth-abundant elements is a p-type semiconductor with a narrow band gap of ~1.9 eV.^{1–5} CaFe₂O₄ possesses an appropriate conduction band edge position that is more cathodic compared to the potentials of the reduction of H₂O and CO₂.^{1–5} Accordingly, CaFe₂O₄ is a candidate of visible-light-driven photocatalysts for water splitting and CO₂ conversion.^{1–5} Photoelectrochemical hydrogen evolution from water over the CaFe₂O₄ photocathode as well as photocatalytic CO₂ reduction on CaFe₂O₄ powders in water to CH₃OH and HCHO has already been demonstrated in the 1980s^{1,2} and 1990s,³ respectively. Another calcium iron oxide, brownmillerite Ca₂Fe₂O₅, is also a p-type semiconductor with a band gap of 2.0–2.2 eV.⁶ Multifunctional Ca₂Fe₂O₅ is the potential metal oxide for energy and environmental applications.^{6–10} In addition to photocatalytic degradation of organic pollutants,^{7–10} Ca₂Fe₂O₅ has been demonstrated to be a photocathode for photoelectrochemical water splitting.⁶ Moreover, nanostructured brownmillerite Ca₂Fe₂O₅ exhibits rapid and reversible CO₂ capture ability.¹¹ However, the visible-light-driven calcium iron oxide photocatalysts possess short carrier diffusion lengths,

which restricts the applications in photocatalytic water splitting and environmental remediation.^{12,13} Formation of a nanostructured heterojunction is a commonly used approach to reduce the photocharge recombination in the photocatalysts with a short carrier diffusion length.^{12–16}

Although their good photocatalytic activities have been reported, both CaFe₂O₄-based and Ca₂Fe₂O₅-based photocatalysts have not been previously investigated for photocatalytic CO₂ conversion in a gaseous system. The efficiency of CO₂ photocatalytic conversion is determined by factors of light harvesting, photocharge separation and transport, as well as CO₂ adsorption and surface reaction taking place on the photocatalysts.^{17,18} Moreover, the direct reduction of water to hydrogen by photoelectrons is a competitive reaction with the photocatalytic conversion of CO₂ in the presence of

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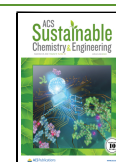


Table 1. DFT Calculated Adsorption Energy of CO₂ at Metal Sites of Mg₂Fe₂O₅, CaFe₂O₄, and Ca₂Fe₂O₅

<i>E</i> _{ad} (eV)	Mg ₂ Fe ₂ O ₅ (001)		CaFe ₂ O ₄ (001)		Ca ₂ Fe ₂ O ₅ (001)		Ca ₂ Fe ₂ O ₅ (010)	
	Mg	Fe	Ca	Fe	Ca	Fe	Ca	Fe
<i>E</i> _{ad} (eV)	−0.473	−3.184	−0.452	−0.519	−0.272	−0.532	1.52	N/A

water.^{17–19} Therefore, improvement of the selectivity of photoelectrons for CO₂ conversion is crucial to enhance the performance of the photocatalyst for CO₂ photocatalytic conversion. Xie et al.²⁰ reported that the reduction of water to hydrogen was suppressed and the selectivity of formation of CH₄ in CO₂ photocatalytic conversion was improved by the addition of MgO into the Pt–TiO₂ photocatalyst, resulting from enhancements of both CO₂ chemisorption and electron density on the surface of the MgO–Pt–TiO₂ catalyst. Bi et al.²¹ demonstrated the Li₂TiO₃/TiO₂ heterostructured photocatalyst with ~100% selectivity for CO₂ photoreduction toward CO, which was achieved by increasing the CO₂ adsorption and the photocharge separation with the introduction of Li₂TiO₃. Accordingly, suitable surface sites for the preferential adsorption of CO₂ coupling with the fast transfer of multiple electrons to the adsorbed CO₂ are essential for the efficacious photocatalytic conversion of CO₂.^{17–21}

On the basis of the density functional theory (DFT) calculations, in this work, Ca₂Fe₂O₅ and Mg₂Fe₂O₅ nanoparticles (NPs) are selected to be the CO₂ adsorption sites of the CaFe₂O₄-based heterostructure for photocatalytic CO₂ conversion. The porous CaFe₂O₄-based heterostructure with the incorporation of Ca₂Fe₂O₅ and Mg₂Fe₂O₅ nanoparticles (NPs) is synthesized using the solution combustion method. Compared to the pristine porous CaFe₂O₄ network, the Mg₂Fe₂O₅ NP-decorated CaFe₂O₄–Ca₂Fe₂O₅ heterostructure demonstrates significantly improved photocatalytic activity for CO₂ and H₂O conversion in a gaseous system to CH₄ and CH₃CHO without H₂ formation. The experimental and DFT calculation results indicate that the CO₂ adsorption, charge separation, and surface CO₂ conversion efficiencies are all enhanced with the addition of Ca₂Fe₂O₅ and Mg₂Fe₂O₅ into the CaFe₂O₄ porous network, resulting in the boosted photocatalytic activity for CO₂ conversion in the Mg₂Fe₂O₅ NP-decorated CaFe₂O₄–Ca₂Fe₂O₅ heterostructure compared to the pristine porous CaFe₂O₄ network.

EXPERIMENTAL SECTION

CaFe₂O₄-based powders were synthesized by the solution combustion method^{22,23} using glycine as the fuel. To prepare the precursor solution for CaFe₂O₄ powders, 1 mmol of Ca(NO₃)₂, 2 mmol of Fe(NO₃)₃, and 2.7 mmol of glycine were dissolved in 10 mL of water. The precursor solution was heated at 85 °C for ~1 h to vaporize water molecules. As soon as the solution became gel-like, the temperature was elevated to 350 °C to trigger the self-propagating combustion. The obtained fluffy powders were calcinated at 850 °C in air for 2 h. For the synthesis of CaFe₂O₄-based heterostructure powders, the aforementioned procedure was conducted except that an additional 0.03 mmol of Mg(NO₃)₂ was added to the 10 mL of aqueous precursor solution.

The morphologies of the CaFe₂O₄-based powders were examined by a scanning electron microscope (SEM, Hitachi SU8010). Crystal structures of the powders were inspected with an X-ray diffractometer (XRD, Rigaku D/MAX-2000) and a transmission electron microscope (TEM, FEI E.O Tecnai F20 G2 MAT S-TWIN). Diffuse reflectance spectra (DRS) of the powders were measured with a UV–vis–IR spectrophotometer (JASCO V-670) with an integrating sphere assembly.

Photocatalytic conversion of CO₂ (99.999%) over the CaFe₂O₄-based powder photocatalysts was carried out at 25 °C in a steady-state gaseous system with a pressure of 1 atm and a CO₂ flow rate of 1 sccm in the presence of H₂O vapor under AM 1.5G simulated sunlight at 100 mW cm^{−2} (100 W, model 94011A, Oriel). A gas chromatograph (GC, Shimadzu, Tracer, GC-2010 Plus) equipped with a barrier flame ionization detector (FID) and an ionization discharge (BID) detector was employed to analyze the gas products. The schematic diagram of the system for photocatalytic CO₂ conversion is illustrated in Figure S1; 10 mg of the photocatalyst was evenly distributed on a susceptor in the reactor with a volume of 142 mL. The volume fraction of H₂O in the CO₂/H₂O gas mixture was estimated to be ~3% based on the saturated vapor pressure of H₂O at 25 °C.

All DFT calculations were carried out by the Vienna Ab initio Simulation Package (VASP) within local density approximation and the plane-wave pseudopotentials.^{24–26} The cutoff energy was set to 520 eV. The convergence tolerances for energy and the residual force on each atom during the structural relaxation were set to 10^{−5} and 0.05 eV, respectively. The 3 × 1 × 1 Mg₂Fe₂O₅ (001) thin-film supercell was relaxed with its bottom layer fixed. Similar operations were applied to the 1 × 1 × 2 CaFe₂O₄ (001), 2 × 2 × 1 Ca₂Fe₂O₅ (001), and 2 × 1 × 1 Ca₂Fe₂O₅ (010) thin-film structures. A 15 Å vacuum layer was added to all of the structures. The activation energy of each step was carried out by the climbing image-nudged elastic band method (CINEB).²⁷

RESULTS AND DISCUSSION

To construct a CaFe₂O₄-based heterostructure for efficient photocatalytic CO₂ conversion, the counterpart of CaFe₂O₄ with the same orthorhombic crystal structure was first evaluated regarding the CO₂ adsorption energy. The adsorption energies of CO₂ on the surfaces of orthorhombic Mg₂Fe₂O₅, Ca₂Fe₂O₅, and CaFe₂O₄ were therefore calculated by DFT using eq 1

$$E_{\text{ad}} = E_{\text{catalyst} + \text{CO}_2(\text{ad})} - E_{\text{catalyst}} - E_{\text{CO}_2} \quad (1)$$

where *E*_{ad} is the adsorption energy of CO₂ on the catalyst surface, *E*_{catalyst + CO₂} denotes the energy of the total system after the adsorption of CO₂ on the catalyst, and *E*_{catalyst} and *E*_{CO₂} are the energies of the catalyst surface and CO₂, respectively.²⁸ A negative value of adsorption energy indicates that CO₂ molecules adsorb on the surface of the catalyst rather than leave the surface. The surface with a more negative adsorption energy, which possesses a stronger interaction with CO₂, is the preferential site for CO₂ adsorption. DFT calculations show that the CO₂ molecule leaves the Fe site of the Ca₂Fe₂O₅ (010) surface during the structure optimization and the Ca site of the Ca₂Fe₂O₅ (010) exhibits a positive value of the adsorption energy for CO₂. This reflects that the Ca₂Fe₂O₅ (010) surface is not suitable for CO₂ reduction. As listed in Table 1, the adsorption energies of CO₂ at the metal sites of Mg₂Fe₂O₅ (001), CaFe₂O₄ (001), and Ca₂Fe₂O₅ (001) surfaces are negative. It reveals that the Fe site of each oxide (001) surface has a stronger interaction with CO₂ than the other metal sites. Moreover, compared to the Fe site of the CaFe₂O₄ (001) surface, the Fe sites of Mg₂Fe₂O₅ (001) and Ca₂Fe₂O₅ (001) surfaces exhibit more negative adsorption

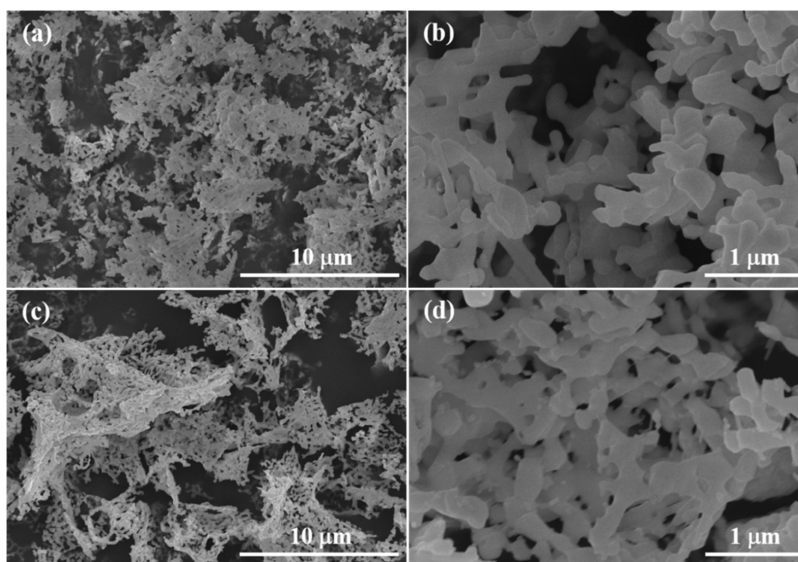


Figure 1. SEM images of (a, b) CFO and (c, d) CMFO synthesized using the solution combustion method.

energy for CO_2 , suggesting that the CO_2 adsorption efficiency will be significantly improved with the incorporation of $\text{Mg}_2\text{Fe}_2\text{O}_5$ and $\text{Ca}_2\text{Fe}_2\text{O}_5$ with CaFe_2O_4 . Due to its strongly negative adsorption energy, the Fe site of the $\text{Mg}_2\text{Fe}_2\text{O}_5$ (001) surface possesses a significant advantage in catching CO_2 compared with all of the other metal sites listed in Table 1. On the basis of DFT calculations, $\text{Mg}_2\text{Fe}_2\text{O}_5$ and $\text{Ca}_2\text{Fe}_2\text{O}_5$ are the potential counterparts of CaFe_2O_4 for the formation of the heterostructure for photocatalytic CO_2 conversion.

In this work, CaFe_2O_4 -based powders were synthesized by the solution combustion method using the fuel of glycine coupling with the aqueous solutions of metal nitride precursors followed by the post-annealing at 850°C in air. For simplicity, the powders produced by the precursors of calcium nitride/ferrite nitride and calcium nitride/magnesium nitrate/ferrite nitride are named CFO and CMFO, respectively. The SEM images of CFO and CMFO are, respectively, shown in Figure 1a–d. They reveal that the aggregated powders with porous networks were produced by the solution combustion method. As displayed in Figure 1b, CFO is composed of striplike powders interconnecting to the porous networks. On the other hand, with the addition of magnesium nitrate in the aqueous precursor solution for CMFO, Figure 1d shows that the tiny particles were simultaneously grown on the striplike powders of the porous networks.

Crystal structures of CFO and CMFO were first characterized by XRD. The XRD patterns of CFO and CMFO are shown in Figure 2. For comparison, the standard X-ray diffractions of orthorhombic CaFe_2O_4 (00-032-0168), orthorhombic $\text{Ca}_2\text{Fe}_2\text{O}_5$ (00-047-1744), cubic MgFe_2O_4 (01-089-6187), and orthorhombic $\text{Mg}_2\text{Fe}_2\text{O}_5$ (01-085-6464) are also illustrated in this figure. The diffraction peaks appearing in the XRD pattern of CFO all correspond to those of orthorhombic CaFe_2O_4 , as shown in Figure 2. It indicates the successful synthesis of orthorhombic CaFe_2O_4 powders using the solution combustion method. In addition to the diffraction peaks pertaining to orthorhombic CaFe_2O_4 , the diffraction peaks at $2\theta = 56.9$, 62.4 , and 62.6° are present in the pattern of CMFO, which can be assigned to the diffraction peaks of $\text{Mg}_2\text{Fe}_2\text{O}_5$ (150), $\text{Ca}_2\text{Fe}_2\text{O}_5$ (351), and $\text{Ca}_2\text{Fe}_2\text{O}_5$ (182), respectively. Accordingly, the result reveals that CMFO

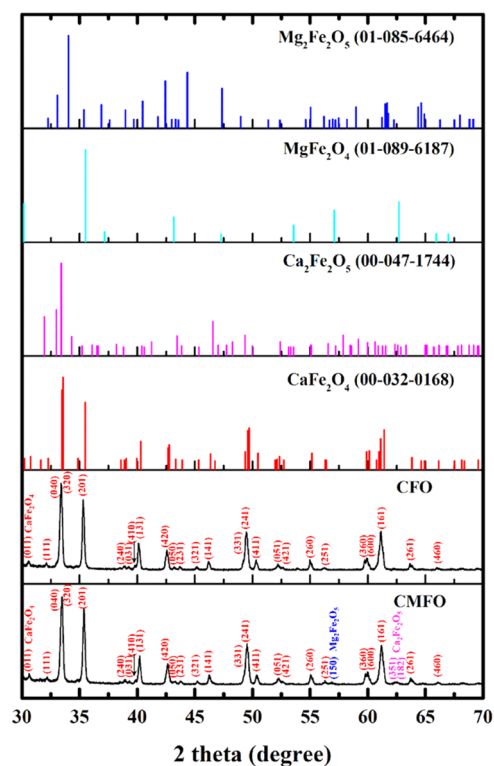


Figure 2. XRD patterns of CFO and CMFO. Standard X-ray diffraction patterns of orthorhombic CaFe_2O_4 , orthorhombic $\text{Ca}_2\text{Fe}_2\text{O}_5$, cubic MgFe_2O_4 , and orthorhombic $\text{Mg}_2\text{Fe}_2\text{O}_5$ are also illustrated.

exhibits the heterostructure composed of crystalline CaFe_2O_4 , $\text{Mg}_2\text{Fe}_2\text{O}_5$, and $\text{Ca}_2\text{Fe}_2\text{O}_5$ powders.

The microstructures of CFO and CMFO were further examined using TEM. Figure 3a shows the TEM image of CFO. The TEM image of CFO clearly displays that the porous networks of CFO are composed of striplike powders, which is consistent with the SEM images shown in Figure 1b. Figure 3b and the inset show the high-resolution (HR)TEM image and the corresponding selected area electron diffraction (SAED) pattern of CFO, respectively. The lattice spacings of 0.29 and

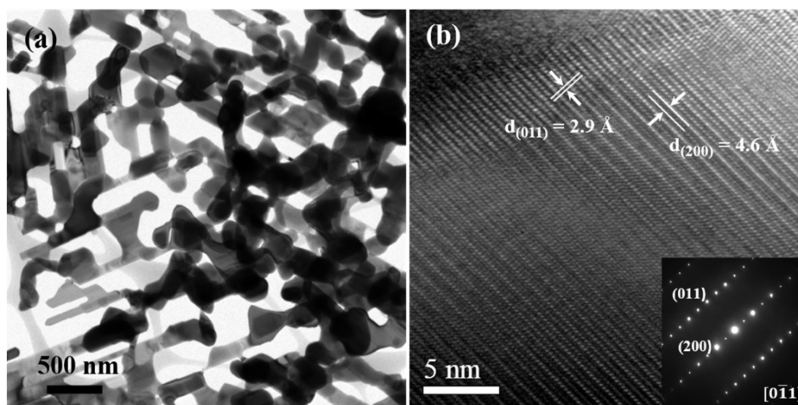


Figure 3. (a) TEM and (b) HRTEM images of CFO. The inset in panel (b) is the corresponding SAED pattern.

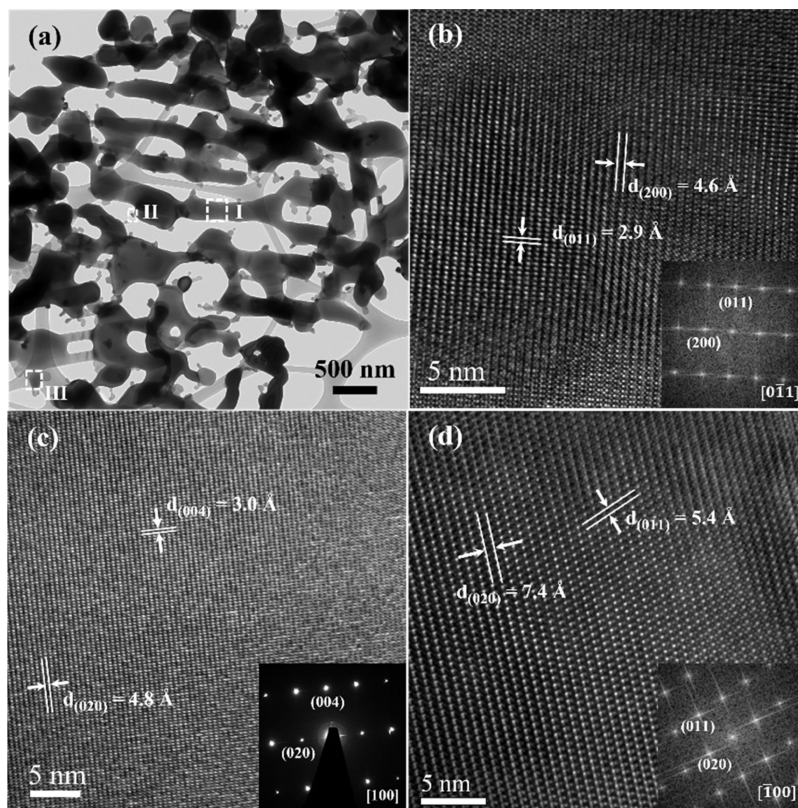


Figure 4. (a) TEM image of CMFO. (b, d) HRTEM images and the corresponding FFT patterns (inset) of regions I and III denoted in panel (a), respectively. (c) HRTEM image and the corresponding SAED pattern (inset) of region II.

0.46 nm denoted in the HRTEM image are consistent with the d -spacings of the (011) and (200) crystal planes of orthorhombic CaFe_2O_4 . The diffraction spots pertaining to the (011) and (200) diffractions of orthorhombic CaFe_2O_4 are shown in the SAED pattern. Accordingly, TEM characterizations confirm the formation of striplike orthorhombic CaFe_2O_4 powders using the solution combustion method.

As shown in Figure 4a, three types of powders with distinct morphologies appear in the TEM image of CMFO. The porous networks of CMFO are mainly assembled by the striplike powders as well. Nevertheless, the surfaces of the striplike powders are partially covered by nanoparticles with a size of ~ 50 nm, as the one denoted in region II. In addition, nanoparticles with a size of ~ 150 nm also exist in the porous network as denoted in region III of Figure 4a. The energy-

dispersive X-ray spectra (EDS) taken from regions I to III of Figure 4a are shown in Figure S2a–c, respectively. Figure S2a,c reveals that metallic elements of Ca and Fe exist in the striplike powder and the nanoparticles with a size of ~ 150 nm. Semiquantitative analyses of the elemental compositions of the striplike powder and the nanoparticles were conducted using EDS. As listed in Table S1, the atomic ratios of Ca and Fe in regions I and III are 1:2 and 1:1, respectively. It is suggested that the striplike powder and the nanoparticles, respectively, exhibit CaFe_2O_4 and $\text{Ca}_2\text{Fe}_2\text{O}_5$ phases. On the other hand, as shown in Figure S2b, Mg and Fe are the major metallic elements in the smaller nanoparticles of CMFO.

The HRTEM characterizations of the three types of powders assembling the heterostructures of CMFO, as denoted in regions I–III of Figure 4a, are shown in Figure 4b–d,

respectively. As illustrated in Figure 4b, the lattice fringes in the HRTEM image and the diffraction spots in the fast Fourier transform (FFT) pattern pertain to the (011) and (200) planes of the orthorhombic CaFe_2O_4 phase. It is consistent with the EDS results shown in Figure S2a and Table S1 that the atomic ratio of Ca and Fe in region I is 1:2. Accordingly, the striplike powders in CMFO exhibit the same crystal structure of orthorhombic CaFe_2O_4 as those in CFO. The EDS spectrum in Figure S2b indicates the existence of Mg and Fe elements in the nanoparticles of region II. The lattice spacings of 0.30 and 0.48 nm denoted in the HRTEM image of the nanoparticles, as shown in Figure 4c, are consistent with the d -spacings of the (004) and (020) crystal planes of orthorhombic $\text{Mg}_2\text{Fe}_2\text{O}_5$. The diffraction spots attributed to the (004) and (020) diffractions of orthorhombic $\text{Mg}_2\text{Fe}_2\text{O}_5$ are shown in the SAED pattern. Characterizations shown in Figure 4c confirm that the nanoparticle with the size of ~ 50 nm in the heterostructure of CMFO exhibits the crystal structure of orthorhombic $\text{Mg}_2\text{Fe}_2\text{O}_5$. On the other hand, the HRTEM image in Figure 4d (the region III of Figure 4a) reveals the lattice fringes with spacings of 0.74 and 0.54 nm, which are consistent with those of the (020) and (011) planes of the orthorhombic $\text{Ca}_2\text{Fe}_2\text{O}_5$ phase. The diffraction spots in the FFT pattern of this region, as illustrated in the inset of Figure 4d, are ascribed to the (020) and (011) planes of orthorhombic $\text{Ca}_2\text{Fe}_2\text{O}_5$. The TEM results confirm the semiquantitative analyses of EDS shown in Figure S2c and Table S1 that the nanoparticle with the size of ~ 150 nm exhibits the crystal structure of orthorhombic $\text{Ca}_2\text{Fe}_2\text{O}_5$. On the basis of the results of XRD, EDS, and TEM characterizations, it can be concluded that orthorhombic $\text{Ca}_2\text{Fe}_2\text{O}_5$ nanoparticles distribute on the striplike orthorhombic CaFe_2O_4 to form the heterojunction. Moreover, the orthorhombic $\text{Mg}_2\text{Fe}_2\text{O}_5$ nanoparticles with the size of ~ 50 nm partially cover the orthorhombic heterostructure network of CMFO.

Figure 5 shows the UV–vis diffuse reflectance spectra of CFO and CMFO porous networks. The absorption edge of

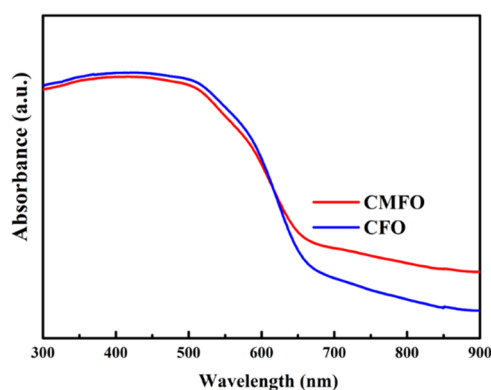


Figure 5. UV–vis diffuse reflectance spectra of CFO and CMFO.

CFO is at ~ 645 nm, corresponding to a direct band gap of 1.92 eV. It is consistent with the band gap of CaFe_2O_4 reported in the literature.^{1–5} CMFO composed of $\text{CaFe}_2\text{O}_4/\text{Mg}_2\text{Fe}_2\text{O}_5/\text{Ca}_2\text{Fe}_2\text{O}_5$ heterostructures exhibits the primary absorption edge at ~ 645 nm as well. It is attributed to the absorption characteristic of CaFe_2O_4 skeletons in the heterostructure network of CMFO. Nevertheless, as shown in the diffuse reflectance spectra, CMFO has higher light-

harvesting ability compared to CFO at $\lambda > 645$ nm. We suggest that the $\text{Mg}_2\text{Fe}_2\text{O}_5$ nanoparticles with a band gap of ~ 0.6 eV²⁹ contribute to the absorption at longer wavelengths although the amount of $\text{Mg}_2\text{Fe}_2\text{O}_5$ nanoparticles is much less than that of the CaFe_2O_4 skeletons in the network. On the other hand, the absorption edge of the $\text{Ca}_2\text{Fe}_2\text{O}_5$ nanoparticles in the heterostructures is absent in the diffuse reflectance spectrum of CMFO because of its larger band gap of 2.2 eV.^{6–10}

Photocatalytic conversion of CO_2 and H_2O over CFO and CMFO photocatalysts was examined in a steady-state gaseous system under AM 1.5G simulated sunlight at 100 mW cm^{-2} . The products were monitored using a GC equipped with an FID and a BID detector. There are no detectable H_2 , CO, and hydrocarbon species during the blank experiment in the absence of CO_2 . The results of photocatalytic CO_2 conversion using CFO and CMFO as photocatalysts are shown in Figure 6a. When CFO is employed as the photocatalyst, no detectable

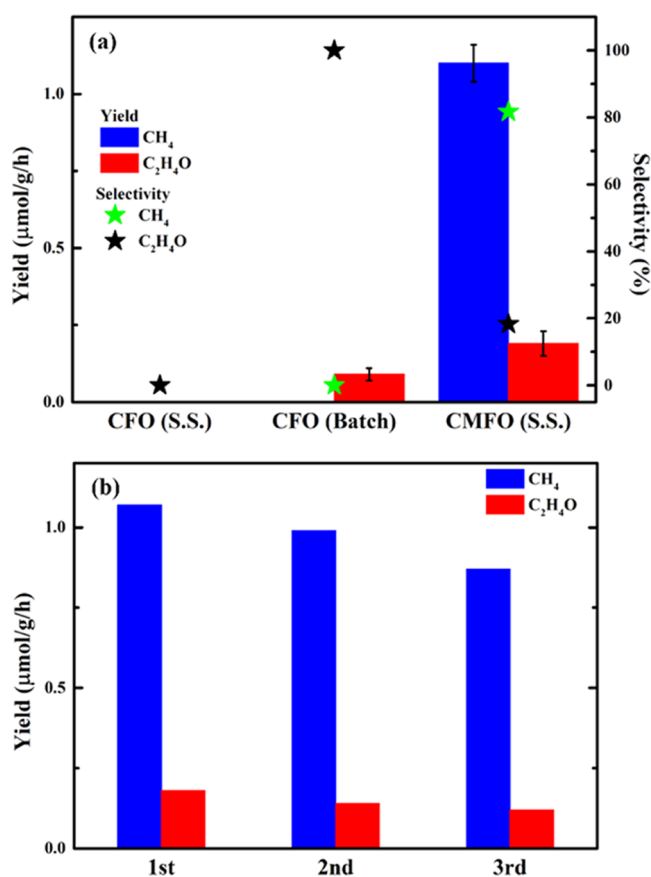


Figure 6. (a) Formation rates (R) of methane and acetaldehyde as well as the corresponding selectivity over CFO and CMFO in the steady-state (SS) and batch gaseous systems for photocatalytic CO_2 conversion. Selectivities for CH_4 and $\text{C}_2\text{H}_4\text{O}$ are calculated by the relations of $8R_{\text{CH}_4}/(8R_{\text{CH}_4} + 10R_{\text{C}_2\text{H}_4\text{O}})$ and $10R_{\text{C}_2\text{H}_4\text{O}}/(8R_{\text{CH}_4} + 10R_{\text{C}_2\text{H}_4\text{O}})$, respectively. (b) Cycling stability evaluation of CMFO for photocatalytic CO_2 conversion.

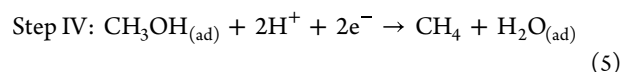
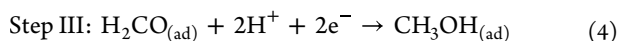
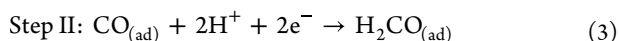
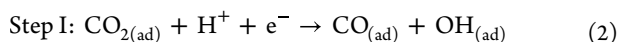
product is obtained during the 10 h measurement in the steady-state system. By contrast, CH_4 and CH_3CHO are produced from the photocatalytic CO_2 conversion process using the CMFO photocatalyst in the steady-state system. It is worth addressing that H_2 evolution through proton reduction is the competition reaction of CO_2 photoconversion in the presence of H_2O as the hole scavenger. Nevertheless, there is

no H₂ detected during the 10 h measurement, indicating the 100% selectivity of CO₂ photoconversion using the CMFO photocatalyst. The formation rates and the selectivities of CH₄ and CH₃CHO through CMFO after a 10 h photocatalytic CO₂ conversion process are shown in Figure 6a. The formation rates of CH₄ and CH₃CHO from the photocatalytic CO₂ conversion process over CMFO are 1.1 and 0.2 μmol g⁻¹ h⁻¹, respectively. 8 and 10 electrons are correspondingly needed for the reduction of CO₂ to CH₄ and CH₃CHO in the presence of H₂O.^{17,18} Accordingly, the selectivities of CH₄ and CH₃CHO are 81.7 and 18.3%, respectively, as shown in Figure 6a. The photocatalytic activity of CFO for CO₂ conversion was further examined in a batch system for 10 h with a pressure of 1 atm in the presence of H₂O vapor. In this case, CH₃CHO with the formation rate of 0.09 μmol g⁻¹ h⁻¹ is the sole product obtained from the batch process of photocatalytic CO₂ conversion, as shown in Figure 6a. CFO exhibits the photocatalytic ability for CO₂ conversion in the batch system. Nevertheless, the formation rate of the hydrocarbon over CFO in the batch system is still considerably inferior to that over CMFO in the steady-state system.

Selective photocatalytic CO₂ conversion to acetaldehyde (CH₃CHO) has been demonstrated using SnS₂-based photocatalysts,^{19,30} although C1 hydrocarbons, e.g., CH₄ and CO, are the common products for the photocatalytic CO₂ conversion process.^{17,18} In addition to CH₄, formation of the C2 hydrocarbon is achieved in this work by employing the Mg₂Fe₂O₅ NP-decorated CaFe₂O₄-Ca₂Fe₂O₅ heterostructure for CO₂ conversion. Moreover, there is no H₂ detected in this process. It reflects that this calcium iron oxide-based heterojunction is a promising photocatalyst for photocatalytic CO₂ conversion. The stability of photocatalytic CO₂ conversion over CMFO was further examined for the 10 h process with three cycles in the steady-state gaseous system under AM 1.5G simulated sunlight at 100 mW cm⁻². Figure 6b displays the formation rates of CH₄ and CH₃CHO through CMFO for each cycle after the 10 h photocatalytic CO₂ conversion process. The formation rates of both products are gradually degraded, revealing that the CMFO photocatalyst maintains good stability for CO₂ conversion enduring three cycles for 30 h.

The results of photocatalytic CO₂ conversion over CFO and CMFO indicate that the construction of heterostructures by the additions of Mg₂Fe₂O₅ and Ca₂Fe₂O₅ nanoparticles on the surface of CaFe₂O₄ skeletons in the network significantly improves the photocatalytic activity of the CaFe₂O₄ porous network photocatalyst for CO₂ photoconversion. As listed in Table 1, DFT calculations reveal that the CO₂ adsorption energies on both Ca₂Fe₂O₅ and Mg₂Fe₂O₅ surfaces are stronger than that on the CaFe₂O₄ surface. Nevertheless, the active site of CMFO needs to be further determined for the understanding of the mechanism for photocatalytic CO₂ conversion.

To study the active site of CMFO for the photocatalytic conversion of CO₂ and H₂O to CH₄, DFT calculations on the formaldehyde pathway³¹



taking place at the metal sites of Mg₂Fe₂O₅ (001), Ca₂Fe₂O₅ (001), and CaFe₂O₄ (001) were performed in this work. The adsorption energies of the reactant for each step at these metal sites were first calculated. Apart from the reactant of step III at the Mg site of Mg₂Fe₂O₅ (001), the adsorption energies with negative values were obtained for the reactants at the rest of the metal sites on CMFO. A H₂CO molecule cannot adsorb at the Mg site of Mg₂Fe₂O₅ (001), reflecting that the Mg site is not a possible active site for the conversion of CO₂ and H₂O to CH₄. The activation energies for each step of the pathway occurring at the possible active sites were further calculated by the CINEB method after all of the intermediates were relaxed on the surface of metal oxides. The potential energy surfaces of the Fe site of Mg₂Fe₂O₅ (001), the Fe and Ca sites of CaFe₂O₄ (001), and the Ca site of Ca₂Fe₂O₅ (001) for the conversion of CO₂ and H₂O to CH₄ are shown in Figure S3a–d, respectively. The Fe sites of both Mg₂Fe₂O₅ (001) and CaFe₂O₄ (001) exhibit high activation energies of 3.46 and 4.38 eV for step I, respectively, as illustrated in Figure S3a,b. In addition, Figure S3d reveals that the energy barrier for step I at the Ca site of Ca₂Fe₂O₅ (001) is up to 11.1 eV. At the Ca site of CaFe₂O₄ (001), as shown in Figure S3c, the energy barrier of 3.48 eV is observed for step III although the activation energy for step I is relatively low. The potential energy surface of the Fe site of Ca₂Fe₂O₅ (001) for the conversion of CO₂ and H₂O to CH₄ is displayed in Figure 7. It is evidently

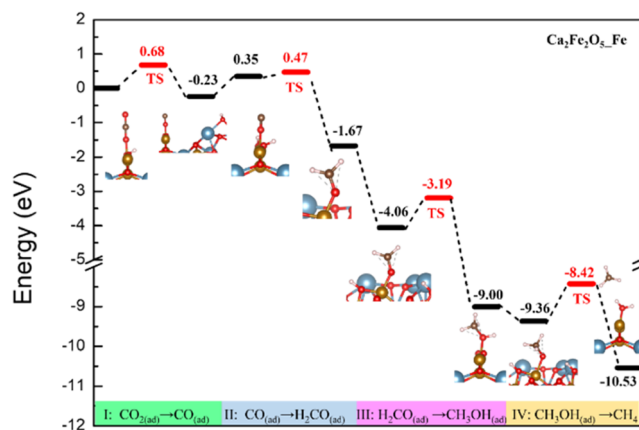


Figure 7. DFT calculated potential energy surface of the Fe site of Ca₂Fe₂O₅ (001) for the conversion of CO₂ and H₂O to CH₄.

demonstrated that the formaldehyde pathway is thermodynamically feasible on the Fe site of Ca₂Fe₂O₅ (001). Step IV, which exhibits the largest activation energy of 0.94 eV among the sequential steps, is the rate-determining step of the reaction pathway taking place at the Fe site of Ca₂Fe₂O₅ (001). The activation energy of the rate-determining step for the conversion of CO₂ and H₂O to CH₄ occurring at the Fe site of Ca₂Fe₂O₅ (001) is much lower than those at the other metal sites on CMFO. Accordingly, the DFT results indicate that the Fe site of Ca₂Fe₂O₅ (001) is the active site of CMFO for the conversion of CO₂ and H₂O to CH₄.

As shown in Figure 6a, the photocatalytic activity for CO₂ conversion is significantly improved by the additions of Mg₂Fe₂O₅ and Ca₂Fe₂O₅ nanoparticles on the surface of CaFe₂O₄ skeletons in the network. DFT calculations indicate

that the CO_2 adsorption efficiency can be considerably enhanced by the incorporation of $\text{Mg}_2\text{Fe}_2\text{O}_5$ and $\text{Ca}_2\text{Fe}_2\text{O}_5$ with CaFe_2O_4 , as listed in Table 1. Figure S3 reveals that compared to the other metal sites, the activation energy for the conversion of CO_2 to CH_4 taking place on the metal site of $\text{Mg}_2\text{Fe}_2\text{O}_5$ is rather high. Moreover, due to its small band gap of 0.6 eV,²⁹ the band edge positions of $\text{Mg}_2\text{Fe}_2\text{O}_5$ will not be suitable for the redox reactions of the photocatalytic CO_2 conversion occurring on CMFO. Accordingly, the reaction center of the CMFO photocatalyst for CO_2 conversion is located at the heterojunction of CaFe_2O_4 and $\text{Ca}_2\text{Fe}_2\text{O}_5$. A staggered band structure is suggested to be constructed in the CaFe_2O_4 – $\text{Ca}_2\text{Fe}_2\text{O}_5$ heterojunction.^{4,6,14} On the basis of the DFT calculations shown in Figures 7 and S3, compared to the metal sites of CaFe_2O_4 , the Fe site of $\text{Ca}_2\text{Fe}_2\text{O}_5$ (001) possesses a much lower activation energy for the conversion of CO_2 and H_2O to CH_4 . The Fe site of $\text{Ca}_2\text{Fe}_2\text{O}_5$ (001) is the preferential active site of CMFO for photocatalytic CO_2 conversion.

We suggest that the staggered CaFe_2O_4 – $\text{Ca}_2\text{Fe}_2\text{O}_5$ heterojunction experiences type-II charge transfer dynamics, as illustrated in Figure 8. Compared to pristine CaFe_2O_4 ,

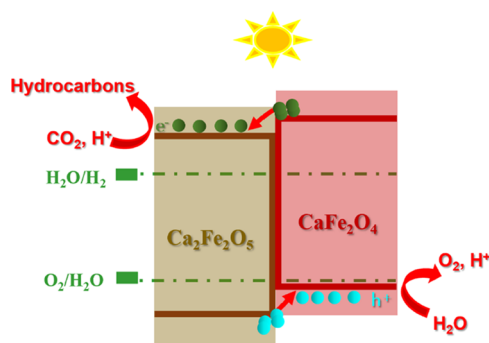


Figure 8. Proposed band diagram of the CaFe_2O_4 – $\text{Ca}_2\text{Fe}_2\text{O}_5$ heterojunction and the corresponding photocatalytic reaction mechanism for CO_2 conversion.

photocharge separation is improved due to the built-in field in the CaFe_2O_4 – $\text{Ca}_2\text{Fe}_2\text{O}_5$ heterojunction. Before charge recombination occurs, the photogenerated holes in $\text{Ca}_2\text{Fe}_2\text{O}_5$ rapidly move to CaFe_2O_4 for water oxidation. On the other hand, the photogenerated electrons in CaFe_2O_4 swiftly transfer to $\text{Ca}_2\text{Fe}_2\text{O}_5$, which is the preferential site for CO_2 adsorption as well as for CO_2 conversion to CH_4 on CMFO. Therefore, the formation of the CaFe_2O_4 – $\text{Ca}_2\text{Fe}_2\text{O}_5$ heterojunction in CMFO improves not only the charge separation but also the surface conversion rate, resulting in the boosted photocatalytic activity for CO_2 conversion compared to CFO.

CONCLUSIONS

In this work, a CaFe_2O_4 porous network composed of striplike powders is synthesized using the solution combustion method, which shows limited performance for photocatalytic CO_2 conversion. With the incorporation of $\text{Ca}_2\text{Fe}_2\text{O}_5$ and $\text{Mg}_2\text{Fe}_2\text{O}_5$ NPs into the network, the $\text{Mg}_2\text{Fe}_2\text{O}_5$ NP-decorated CaFe_2O_4 – $\text{Ca}_2\text{Fe}_2\text{O}_5$ heterostructures demonstrated significantly improved photocatalytic activity for CO_2 conversion. DFT calculation results indicate that the CO_2 adsorption energies on both $\text{Ca}_2\text{Fe}_2\text{O}_5$ and $\text{Mg}_2\text{Fe}_2\text{O}_5$ surfaces are stronger than that on the CaFe_2O_4 surface. Moreover, compared to CaFe_2O_4 and $\text{Mg}_2\text{Fe}_2\text{O}_5$, $\text{Ca}_2\text{Fe}_2\text{O}_5$ possesses

much lower activation energy for the conversion of CO_2 and H_2O to CH_4 . Type-II charge transfer dynamics takes place in the staggered CaFe_2O_4 – $\text{Ca}_2\text{Fe}_2\text{O}_5$ heterojunction to improve charge separation, which allows the photoelectrons to be well collected on the active site of $\text{Ca}_2\text{Fe}_2\text{O}_5$ for the reduction of CO_2 . Compared to a pristine CaFe_2O_4 porous network, the photocatalytic performance for the conversion of CO_2 over the $\text{Mg}_2\text{Fe}_2\text{O}_5$ NP-decorated CaFe_2O_4 – $\text{Ca}_2\text{Fe}_2\text{O}_5$ heterostructure is boosted by the enhancements of CO_2 adsorption, charge separation, and surface CO_2 conversion efficiencies.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.2c03378>.

EDS spectra of the $\text{Mg}_2\text{Fe}_2\text{O}_5$ NP-decorated CaFe_2O_4 – $\text{Ca}_2\text{Fe}_2\text{O}_5$ heterostructure and the potential energy surfaces of $\text{Mg}_2\text{Fe}_2\text{O}_5$, CaFe_2O_4 , and $\text{Ca}_2\text{Fe}_2\text{O}_5$ for the conversion of CO_2 and H_2O to CH_4 (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval for the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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